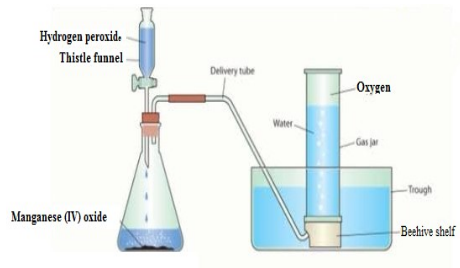
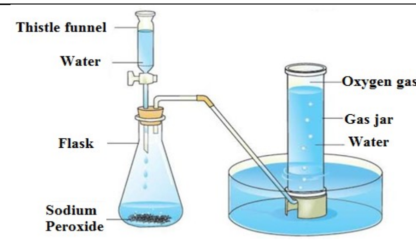
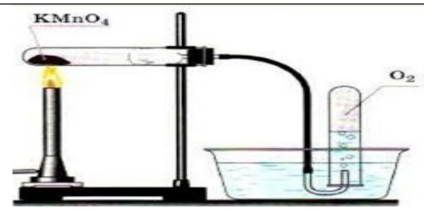
Summarized lab preparation of Gases

GAS	REACTANTS, CONDITIONS & EQUATION	DIAGRAM/SET-UP	DRYING	COLECTION
	Hydrogen peroxide and catalyst manganese (iv) oxide. Hydrogen peroxide $\xrightarrow{MnO_2}$ water + oxygen gas $2H_2O_{2(aq)} \xrightarrow{MnO_2} 2H_2O_{(l)} + O_{2(g)}$			Downward displacement of water/ over water method. Because it is slightly soluble in water.
OXYGEN	Sodium peroxide with water. $2H_2O(l) + 2Na_2O_2(s) \longrightarrow 4NaOH(aq) + O_2(g)$			Downward displacement of water/ over water method. Because it is slightly soluble in water.
	Decomposition of Potassium Manganate (VII) $2KMnO_4(s) \xrightarrow[\boxed{Heat}]{Heat} K_2MNO_4(s) + MnO_2(s) + O_2(g)$ **Test** The gas relights/rekindles a glowing splint.			Downward displacement of water/ over water method. Because it is slightly soluble in water.

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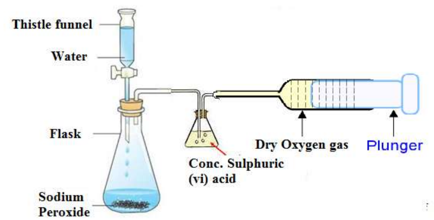
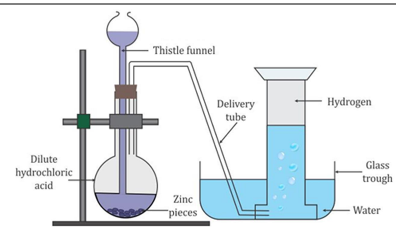
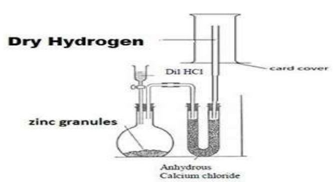
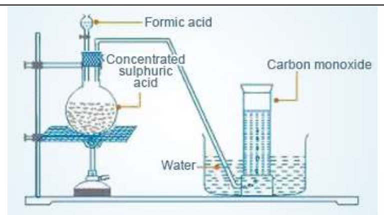
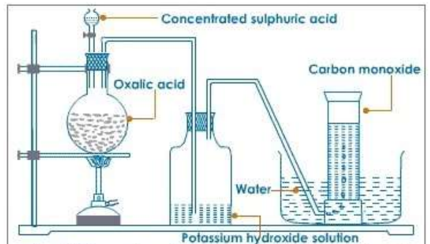
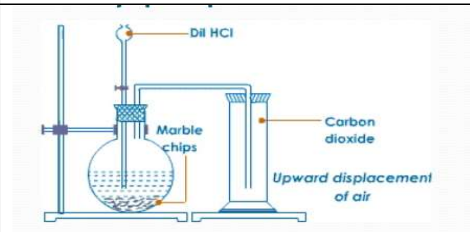
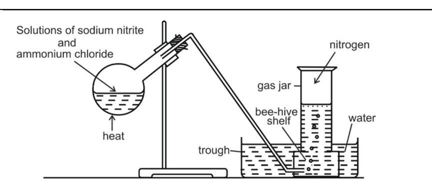
	DRY GAS		**Concentrated sulphuric (vi) acid.** **or** **Anhydrous calcium chloride.** **or** **Calcium Oxide.**	**Using syringe** **Because it has the same density as air.**
Hydrogen	Zinc granules and dilute hydrochloric acid/ dilute sulphuric (vi) acid. $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$ $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$			**Downward displacement of water/ over water method.** **Because it is slightly soluble in water.**
	Test **It puts off a burning splint with a pop sound.**		**Concentrated sulphuric (vi) acid.** **or** **Anhydrous calcium chloride.**	**Upward delivery of air/downward displacement of air.** **Because it is less dense than air.**
Carbon (II) Oxide	Dehydration of methanoic acid /Formic acid (HCOOH) OR Ethan-1,2-dioic acid/ oxalic acid(HOOCCOOH) using concentrated sulphuric (VI) acid. Heat is necessary. $H_2CO_2(s) \xrightarrow{\text{Conc } H_2SO_4} CO(g) + H_2O(l)$		**Concentrated sulphuric (vi) acid.** **or** **Anhydrous calcium chloride.**	**Downward displacement of water/ over water method.** **Because it is slightly soluble in water.** **Downward delivery/upward displacement method when**

			or Calcium Oxide.	needed dry. Because it is more dense than air.
	$H_2C_2O_4(s) \xrightarrow{Conc.H_2SO_4} CO(g) + CO_2(g) + H_2O(l)$ **Potassium hydroxide or sodium hydroxide solution is used to absorb/remove carbon (iv) oxide during the reaction.** **TEST** **Burns in air with a blue flame producing a gas that forms a white precipitate with calcium hydroxide.**	 *Fig. 11.2 Prepartion of carbon monoxide from oxalic acid*		**Downward displacement of water/ over water method.** **Because it is slightly soluble in water.** **Downward delivery/upward displacement method when needed dry.** **Because it is more dense than air.**
Carbon (iv) oxide	Reaction of marble chips/calcium carbonate ($CaCO_3$) **or** sodium hydrogen carbonate ($NaHCO_3$) with dilute hydrochloric acid. $CaCO_3(s) + HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$ **TEST** **It forms a white precipitate when bubbled in calcium hydroxide solution**		**Concentrated sulphuric (vi) acid.** **or** **Anhydrous calcium chloride.**	**Downward delivery/upward displacement method when needed dry.** **Because it is more dense than air.** **Note:** **the gas cannot be collected by over water method because it is highly soluble in water forming a weak carbonic acid.**
Nitrogen	Heating a mixture of ammonium chloride and sodium nitrite. $NaNO_2(aq) + NH_4Cl(aq) \xrightarrow{Heat} NaCl(aq) + H_2O + N_2(g)$		**Concentrated sulphuric (vi) acid.** **or** **Anhydrous calcium chloride.**	**Downward displacement of water/ over water method.** **Because it is slightly soluble in water.** **Upward delivery of air/downward displacement**

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				of air when needed dry. Because it is less dense than air.